Regression exercise

Using kidney data[1]

1. Fit a linear regression as a function of age. Reproduce the plot and table from *CASI* Figure 1.1 and Table 1.1 (you do not need to recreate lowess).

2. Fit a cubic regression as a function of age similar to the *CASI* Figure 1.1. Calculate estimates and standard errors at ages 20, 30, 40, 50, 60, 70, 80. How do the results compare with those in Table 1.1? Create a new plot and table for the cubic regression.

3. Reflect on what insights the different fits provide; why would one prefer one over the other? What does all of this suggest about kidneys?

You can solve these exercises in any way you like, with whatever your preferred programming language is. If you use Python, we recommend using *statsmodels*' linear regression. When loading the data in Python, you can use the following code snippet to load the data into a *Pandas* dataframe:

```
import requests
from io import StringIO
import pandas as pd

url = 'https://hastie.su.domains/CASI_files/DATA/kidney.txt'
raw_data = requests.get(url).text
csv_string_io = StringIO(raw_data)
df = pd.read_csv(csv_string_io, sep=' ')
df.head()
```

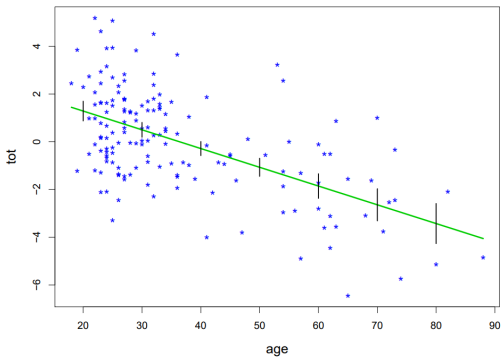


Figure 1.1 Kidney fitness `tot` vs `age` for 157 volunteers. The line is a linear regression fit, showing ±2 standard errors at selected values of `age`.

Table 1.1 *Regression analysis of the kidney data; (1) linear regression estimates; (2) their standard errors; (3)* `lowess` *estimates; (4) their bootstrap standard errors.*

`age`	20	30	40	50	60	70	80
1. linear regression	1.29	.50	−.28	−1.07	−1.86	−2.64	−3.43
2. std error	.21	.15	.15	.19	.26	.34	.42
3. lowess	1.66	.65	−.59	−1.27	−1.91	−2.68	−3.50
4. bootstrap std error	.71	.23	.31	.32	.37	.47	.70

[1] `https://hastie.su.domains/CASI_files/DATA/kidney.txt`